On-Device Speech Recognition on iOS: A Short Overview

Prepared by BeanBao · Nov 2, 2026

1. Classic option: SFSpeechRecognizer (iOS 13+)

Apple documents SFSpeechRecognizer.supportsOnDeviceRecognition as a Boolean showing whether the recognizer can work without network access (iOS 13+). A request's requiresOnDeviceRecognition is honored only if that property is true. Setting it to true keeps audio on the device. Apple warns that on-device requests won't be as accurate. If supportsOnDeviceRecognition is false, recognition requires a network. [4] [3]

Since iOS 17, apps can customize the SFSpeechRecognizer language model for on-device recognition (WWDC23 session 'Customize on-device speech recognition'). [10]

Caveat (community reports, anecdotal, not Apple-confirmed): supportsOnDeviceRecognition returning true may not mean the model is installed. Developers report errors such as 'Failed to access assets', and a forum reply says offline use needs the matching keyboard and dictation language installed. Another thread reports earlier transcription being discarded in on-device mode. Test on real devices for each locale. [2] [5] [6] [7]

2. Newer Apple option: SpeechAnalyzer / SpeechTranscriber (iOS 26+)

Apple's WWDC25 session 277 introduces SpeechAnalyzer as a new speech-to-text API with a new Apple model. It does speech-to-text entirely on device with little code. You add a transcriber module to an analyzer session, feed it audio buffers, and receive text with metadata. AssetInventory manages the language models, and apps can hold only a limited number of languages at a time. Apple's documentation lists SpeechAnalyzer as available from iOS 26.0. [11] [9] [8]

Secondary summary (a third-party copy of Apple content, not apple.com): SpeechAnalyzer is described as better suited to long-form and distant audio than the older API. [12]

3. Third-party option: Whisper via WhisperKit (Core ML)

OpenAI's Whisper is a Transformer model for multilingual speech recognition. Its official repo is Python-based, not a Core ML package. Argmax's open-source Swift package (WhisperKit, now part of argmax-oss-swift) deploys Whisper on device on Apple platforms. It downloads a recommended model for the device if none is specified. Converted Core ML models are hosted on Hugging Face. [14] [13] [15]

Argmax's ICML 2025 abstract reports 0.46s latency and 2.2% WER against server-side systems. This is a vendor-run result and I found no independent replication. Argmax's docs say the advanced streaming algorithm is in its paid Pro SDK. [16] [1]

4. Comparisons and uncertainty

Secondary or vendor comparisons only (not Apple claims). Argmax, which sells a competing SDK, lists Apple SpeechTranscriber at 14.0 error rate and a speed factor of 70. A Kompozy article cites hands-on tests where a SpeechAnalyzer tool transcribed a 34-minute video about 2.2x faster than local Whisper Large V3 Turbo. Methodology was not verified for either, so treat these as indicative. [19] [17]

Gaps: I found no current official list of on-device locales. Older language-count figures, from a 2022 blog, are dated, so check supported locales at runtime. I did not retrieve the full SpeechTranscriber reference page, so its locale support and accuracy are unconfirmed. [18] [8]

5. Practical takeaway

Synthesis of the sources above. For iOS 26+, evaluate SpeechAnalyzer/SpeechTranscriber first, since Apple positions it as the fully on-device path. For older iOS versions, use SFSpeechRecognizer: check supportsOnDeviceRecognition, then set requiresOnDeviceRecognition, and handle missing-asset errors. Consider WhisperKit if you need Whisper models or cross-version consistency, and run your own benchmarks. [11] [4] [13]

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